

BHARATHKUMAR RAJAGOPAL

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🏠 [bharathkumar-rajagopal.github.io](https://github.com/bharathkumar-rajagopal)

🌐 [bharath--r](#)

SKILLS

- [1] **Molecular Biology:** RT-PCR, SDS-PAGE, Western Blotting, DNA Sequencing, Immunoelectrophoresis
- [2] **Analytical Techniques:** HPLC, GC-MS/MS, DLS, NTA, UV-Vis, FTIR, FACS
- [3] **Bioinformatics:** Library Preparation, SWISS-MODEL, Mega X, PyMol, Mothur, QIIME 2
- [4] **Statistical Analysis:** GraphPad Prism
- [5] **Language:** Python (Biopython), R, $\LaTeX$

RESEARCH EXPERIENCE

Independent Researcher | Under Scholarly Mentorship

Jul 2024 – Present

SKILLS: Sample Collection, Mothur, R, Manuscript Preparation

- Research in microbial bioremediation and ecotoxicology under Dr. Boobal Rangaswamy.
- Contributing to scientific manuscripts for peer-reviewed publication.

Dissertation Trainee | CSIR-IITR, Lucknow

Jan 2024 – May 2024

SKILLS: Nanomaterial/pollutant Characterisation, Mammalian Cell Culture, Handling *in vivo* models, Cytoto- & Genotoxicity Assessments

- Conducted *in vitro* toxicity assessment of polystyrene nanoplastics on mammalian lung fibroblast cells
- Characterised nanoplastics using DLS and NTA; performed cytotoxicity assessment (MTT, LDH, NRU, Trypan Blue Dye Exclusion, and ROS generation assays), and Genotoxicity assessment (Comet assay). BALF assessment in murine model (*in vivo*)
- Analysed data via GraphPad; identified ROS-mediated toxicity pathways

Postgraduate Research Assistant | Amity University

Jul 2023 – Dec 2023

SKILLS: Microbial Culture (Isolation, Establishment, Maintenance), Nanodelivery systems, Bioremediation

- Developed pesticide-degrading microbial consortia microbeads.
- Optimised bacterial release efficiency for environmental remediation.

Summer Research Intern | Centre for Bioscience and Nanoscience Research

May 2023 – Jun 2023

SKILLS: Microbial Culture, Sequencing and Sequence Analysis, Nanoparticle Synthesis, Mammalian Cell Culture, Antimicrobial & Anti-cancer Assays

- Isolated Bacteria *Priestia megaterium* and sequenced (GenBank: OR234764.1)
- Biogenically synthesised silver nanoparticles (AgNPs)
- Characterised via UV-Vis, FTIR, and SEM
- Evaluated antimicrobial(*P. aeruginosa*, *S. aureus*, and *K. pneumoniae*) & anticancer(Hep-G2) properties

EDUCATION

Amity University, Noida | MSc Biotechnology | First Class with Distinction | (≥ 97 th percentile in class)

2022 – 2024

- Advanced Microbiology, Advanced Molecular Biology, Applied Genomics & Proteomics, Plant & Animal Biotechnology, Python

PSG College of Arts & Science, Coimbatore | BSc Biotechnology | First Class

2019 – 2022

- Microbiology, Genetics, Environmental Science, Bioinformatics, Analytical Techniques, Computer Applications in Biotechnology

PUBLICATIONS

- [1] **Rajagopal, B.**, Abiraj, V., & Ragunathan, R. (2025). Biogenic synthesis of silver nanoparticles using endophytic bacteria from *Andropogon paniculata*. **In preparation**.
- [2] Sakthivel, I., Shanmugam, L., **Rajagopal, B.**, & Rangaswamy, B. (2025). Integrating kinetic models, gene circuits, and biofilm dynamics. **In Review**- Journal of Microbiological Methods. SSRN Preprint: [doi:10.2139/ssrn.5235712](https://ssrn.com/abstract=5235712)
- [3] Kongaserry, A., Shanmugam, L., **Rajagopal, B.**, & Rangaswamy, B. (2025). Development of *Artocarpus heterophyllus* Lam. nanosuspension. **In Review** - Nanotechnology for Environmental Engineering.

CERTIFICATES

- Introduction to Professional Scientific Communication (NPTEL) Jan 2024 – Feb 2024
- Certificate Course on Bionanotechnology (Association of Indian Biologists) May 2020
- Certificate course in Bioinformatics (PSGCAS) Sep 2019 – May 2020
- Hands-on Training: HPLC & GC-MS/MS (ICFRE-IFGTB) Feb 2020
- Human Molecular Genetics (NPTEL) Jan 2020 – Feb 2020

REFEREES

- **Dr. Boobal Rangaswamy** – Assistant Professor, PSG College of Arts & Science | boobal@psgcas.ac.in
- **Dr. Shashi Sharma** – Assistant Professor, Amity University | ssharma@amity.edu
- **Dr. Ravi Kant Singh** – Professor, Amity University | rksingh1@amity.edu